

Addis Ababa University
School of Pharmacy
Department of Pharmaceutical Chemistry

Course Title: Medicinal Chemistry I

Course Code: Phar 2111

Pre-request: Organic Chemistry

Course EtCTS (Course hour): 7 (189hrs)

Semester II

Instructor: Alemu Tadesse

Co-request: Pharmacology I

Course Objective: After completion of this course, students will be able to understand the drug discovery and designing methods, and also deals with the chemistry of various class of drugs that act on different systems and organs of human body

Course description: Medicinal chemistry is the application of chemistry in the context of human medicine.

Course outline:

1. Introduction to Medicinal Chemistry (12hrs)

- Definition of medicinal chemistry
- Sources of drugs
- Drug targets and drug-target interactions
- Introduction to drug design and discovery
 - The process of drug discovery
 - Analogue design
 - General Process
 - Special Process
 - Structure Activity Relationship (SAR)
 - Pro-drug design
 - Carrier linked pro-drugs
 - Bio-precursor pro-drugs
 - Introduction to CADD
 - QSAR in drug design
 - The partition parameter
 - Electronic parameter
 - Steric parameter
 - Hansch analysis
- Stereochemistry and drug action
- Receptors and drug action
- Drug metabolism

2. Drugs Acting on Autonomic Nervous System (14hrs)

- Cholinergic drugs
 - Cholinergic agonists
 - Cholinergic antagonists
 - ACEI and pesticides

- Adrenergic drugs
 - Adrenergic agonists
 - Adrenergic antagonists
- 3. Diuretics and Cardiovascular Drugs (12hrs)**
 - Diuretics
 - Carbonic anhydrase inhibitors
 - Loop diuretics
 - Thiazide diuretics
 - Potassium sparing diuretics
 - Osmotic diuretics
 - Antianginal agents and vasodilators
 - Antihypertensive agents
 - Drugs used for treatment of congestive heart failure
 - Antiarrhythmic drugs
 - Anti-anemic Agents
 - Drugs for Hypotensive States
 - Antihyperlipidemic drugs
 - Coagulant and anticoagulants
- 4. Oral hypoglycemic Agents (2hrs)**
- 5. Histamine and Antihistaminic Drugs (4hrs)**
 - H₁ – receptor blockers
 - Inhibitors of histamine release
- 6. Analgesics, Antipyretic & anti-inflammatory Agents (4hrs)**
 - Narcotic analgesics and Antagonists
 - Non-narcotic analgesics
 - Drugs used in the treatment of gout
- 7. Drugs Acting on the Respiratory System (4hrs)**
 - Drugs for the Treatment of Asthma
 - Antitussives and Expectorants
 - Nasal Decongestants
- 8. Gastrointestinal and Related Agents (4hrs)**
 - H₂ receptor antagonists and related compounds
 - Proton pump antagonists
 - Miscellaneous gastrointestinal agents
- 9. Pesticides (2hrs)**
 - Classification
 - Specific pesticides
- 10. Antioxidants and Autoxidation (2hrs)**
- 11. Diagnostic Agents (4hrs)**
 - Radio-opaque agents
 - Classification

Books recommended:

1. *Lemke, T.L. and Williams, D.A., Roche, V.F., Zito, W.S. Foye's Principles of Medicinal Chemistry*, 6th. ed. Lippincott Williams & Wilkins, 2008.
2. *Wilson-Gisvold-Doerge; Text book of organic medicinal chemistry and pharmaceutical chemistry*. 12th edn.; Lippincott (USA), 2011.
3. *Gareth Thomas, Medicinal Chemistry An introduction*, 2nd edition, 2007
4. *Rama Rao Nadendla; Principles of Organic Medicinal Chemistry*, 2005.
5. *Burger's Medicinal Chemistry and Drug discovery*, Sixth edition, Volume 1: Drug Discovery; 2003
6. *Burger's Medicinal Chemistry and Drug discovery and drug development*, Sixth edition, Volume 2: Drug Discovery; 2003
7. *Burger's Medicinal Chemistry and Drug discovery*, Sixth edition, Volume 6: Nervous System Agents; 2003
8. *Donald J. Abraham (Ed.). Burger's medicinal Chemistry and Drug Discovery*, 2003, 6th edn., vol1-6, wiley-interscience (USA)
9. *Thomas, L. Lemeke and David, A. Williams. Principle of Medicinal Chemistry*, 2002, 5th edn. A Lea and Febiger book, Williams and Wilkins